

1 Information Theory by Hand

All companion GitHub Code and YouTube videos can be found here:

<https://www.rcalix.com>

Did You Know? Charles Babbage, Computing, and Cryptanalysis

Charles Babbage is famous for designing the Difference Engine and the Analytical Engine, important early steps toward the modern computer. Interestingly, Babbage was also a cryptanalyst. In the nineteenth century, he developed methods for breaking the Vigenère cipher by finding patterns in encrypted messages.

There is some irony in using Babbage as inspiration for a “By Hand” book. Babbage designed machines precisely because he wanted to automate tedious human calculations. In this book, however, we temporarily go in the opposite direction: we perform the calculations by hand so that we can understand what the computer will eventually automate.

In the age of AI, we can increasingly outsource our thinking to machines, but we cannot outsource our understanding.

1.0.1 Problem 1: Entropy from Authentication Logs

A security analyst is monitoring authentication activity on an SSH server. During a short observation period, the following authentication events were recorded:

Event	Authentication Result
1	Success
2	Success
3	Failed
4	Success
5	Failed
6	Success
7	Success
8	Failed
9	Success
10	Failed

The analyst would like to quantify the uncertainty in the authentication results using entropy.

Recall that entropy can be calculated as

$$H(X) = - \sum_x p(x) \log_2 p(x)$$

where $p(x)$ is the probability of observing outcome x .

1. Count the number of successful and failed authentication attempts.
2. Using the observed data, estimate

$$P(\text{Success})$$

and

$$P(\text{Failed}).$$

3. Calculate the entropy of the authentication results by hand.
4. Now consider a second observation period containing 10 authentication attempts, where 5 are successful and 5 are failed. Calculate the entropy for this second observation period.
5. Which observation period has the greater entropy?

6. From a cybersecurity perspective, explain what the difference in entropy tells the security analyst about the authentication activity observed during the two periods.

1.0.2 Problem 2: KL Divergence and Network Traffic Anomalies

A cybersecurity analyst monitors the destination ports used by connections to a server. During a normal observation period, the following 10 connections were recorded:

Connection	Destination Port
1	443
2	443
3	443
4	443
5	443
6	443
7	22
8	22
9	80
10	80

Later, the analyst observes another 10 connections:

Connection	Destination Port
1	443
2	443
3	443
4	22
5	22
6	22
7	22
8	80
9	80
10	80

Treat the first observation period as the normal baseline distribution P and the second observation period as the current distribution Q .

The KL divergence is

$$D_{KL}(P\|Q) = \sum_x P(x) \log_2 \left(\frac{P(x)}{Q(x)} \right).$$

1. Count the number of connections to ports 443, 22, and 80 during the normal observation period.
2. Use these counts to determine the baseline probabilities

$$P(443), \quad P(22), \quad P(80).$$

3. Repeat the process for the current observation period to determine

$$Q(443), \quad Q(22), \quad Q(80).$$

4. Calculate

$$D_{KL}(P\|Q)$$

by hand.

5. Explain what the resulting KL divergence tells the cybersecurity analyst about the current network traffic.

6. Explain how KL divergence could be used as part of an anomaly detection system that continuously compares current network activity with a normal baseline.

1.0.3 Problem 3: From Cybersecurity Data to KL Divergence

A cybersecurity monitoring system records security alerts generated by three different detection rules:

- Authentication Alert
- Port Scanning Alert
- Malware Alert

During a baseline observation period, the following alerts were recorded:

Alert	Alert Type logits vector
2	Authentication
1	Port Scanning
0	Malware

A machine-learning model analyzes one network event and produces the following class scores during the baseline: Authentication = 2, Port Scan = 1, Malware = 0. These are not counts. Instead, they are one vector of logits for just 1 event. Here the system suggests this event is more likely authentication.

During a later observation period, the following alerts were recorded:

Alert	Alert Type logits vector
1	Authentication
2	Port Scanning
0	Malware

These are not counts. Instead, they are one vector of logits for just 1 event. Here the system suggests this event is more likely port scanning.

We will use the number of the logits of each type as a simple score representing the activity observed during each period.

1. Construct the score vector

$$\mathbf{z}_P = [N_{\text{Authentication}}, N_{\text{Scanning}}, N_{\text{Malware}}].$$

2. Repeat the process for the later observation period and construct

$$\mathbf{z}_Q = [N_{\text{Authentication}}, N_{\text{Scanning}}, N_{\text{Malware}}].$$

The values in these vectors are scores, not probability distributions. Machine learning systems commonly use the Softmax function to transform a vector of scores into a probability distribution.

For a score z_i , Softmax is

$$P(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}.$$

For this problem, use

$$e^0 = 1, \quad e^1 \approx 2.718, \quad e^2 \approx 7.389.$$

3. Apply Softmax to the baseline score vector to obtain the probability distribution P .
4. Apply Softmax to the later score vector to obtain the probability distribution Q .
5. Verify that each probability distribution sums to approximately 1.
6. Calculate the KL divergence

$$D_{KL}(P||Q) = \sum_i P_i \ln \left(\frac{P_i}{Q_i} \right).$$

7. Explain what the KL divergence tells the cybersecurity analyst about the change in the distribution of security alerts.
8. PyTorch's KL divergence implementation expects its input in log-probability form. Calculate

$$\ln(P_1), \quad \ln(P_2), \quad \ln(P_3).$$

Explain the relationship between the calculations performed in this problem and the use of `softmax` and `log_softmax` when KL divergence is calculated using PyTorch.

1.0.4 Problem 4: Mutual Information and Cybersecurity Feature Selection

Mutual information can be calculated as

$$I(X; Y) = \sum_{x,y} p(x, y) \log_2 \left(\frac{p(x, y)}{p(x)p(y)} \right)$$

The procedure is:

1. Count the occurrences of each X, Y combination.
2. Calculate the joint probabilities $p(x, y)$.
3. Calculate the marginal probabilities $p(x)$ and $p(y)$.
4. Substitute these probabilities into the mutual information equation.

A cybersecurity analyst is developing a system for identifying malicious login activity. The analyst has collected a small set of previously investigated login attempts. Each login has been labeled as either Normal or Attack.

The analyst is considering whether the feature *Failed Authentication* contains useful information about the security label.

The following observations were collected:

Login	Failed Authentication	Security Label
1	No	Normal
2	No	Normal
3	No	Normal
4	No	Normal
5	Yes	Attack
6	Yes	Attack
7	Yes	Attack
8	No	Attack

Let

$$F = \begin{cases} 0 & \text{No failed authentication} \\ 1 & \text{Failed authentication} \end{cases}$$

and

$$C = \begin{cases} 0 & \text{Normal} \\ 1 & \text{Attack.} \end{cases}$$

Mutual information measures how much information one variable provides about another variable.

For two variables F and C ,

$$I(F; C) = \sum_{f,c} P(f, c) \log_2 \left(\frac{P(f, c)}{P(f)P(c)} \right).$$

The value $P(f, c)$ is the joint probability of observing a particular feature value and class label together.

1. Count the number of Normal and Attack observations and determine

$$P(C = 0) \quad \text{and} \quad P(C = 1).$$

2. Count the number of observations with and without a failed authentication and determine

$$P(F = 0) \quad \text{and} \quad P(F = 1).$$

3. From the original observations, complete the following joint-frequency table:

	$C = 0$	$C = 1$
$F = 0$	?	?
$F = 1$	?	?

4. Convert the joint frequencies into the joint probabilities

$$P(F = 0, C = 0), \quad P(F = 0, C = 1), \quad P(F = 1, C = 0), \quad P(F = 1, C = 1).$$

5. Calculate the mutual information

$$I(F; C)$$

by hand.

When a joint probability is zero, its contribution to the mutual information sum can be treated as zero.

6. If

$$I(F; C) = 0,$$

the failed-authentication feature would provide no information about the security label. Based on the value calculated in this problem, does the feature appear to contain useful information about whether a login is Normal or an Attack?

7. Explain why mutual information could be useful when selecting cybersecurity features for a machine learning or anomaly detection system.

1.0.5 Problem 5: Information Gain from Cybersecurity Data

Information Gain measures how much knowing a feature reduces the uncertainty about an outcome. It is defined as

$$IG(C, F) = H(C) - H(C|F),$$

where C is the outcome and F is a feature.

Consider the following network observations collected by a security analyst:

Connection	Unusual Port	Classification
1	No	Normal
2	No	Normal
3	No	Normal
4	No	Attack
5	Yes	Attack
6	Yes	Attack
7	Yes	Attack
8	Yes	Normal

Let C represent the Classification and F represent whether an Unusual Port was observed.

Recall that entropy is

$$H(C) = - \sum_c p(c) \log_2 p(c).$$

1. Count the number of Normal and Attack connections and determine their probabilities.
2. Calculate the entropy of the classifications,

$$H(C).$$

3. Separate the data into connections where Unusual Port = No and Unusual Port = Yes. For each group, determine the probabilities of Normal and Attack and calculate its entropy.
4. Calculate the conditional entropy

$$H(C|F) = p(F = \text{No})H(C|F = \text{No}) + p(F = \text{Yes})H(C|F = \text{Yes}).$$

5. Calculate the Information Gain

$$IG(C, F) = H(C) - H(C|F).$$

6. Explain what the Information Gain tells the security analyst about the usefulness of the Unusual Port feature.

1.0.6 Problem 6: Conditional Entropy and Security Alerts

Conditional Entropy measures how much uncertainty remains about one variable after another variable is known. It is defined as

$$H(C|F) = \sum_f p(f)H(C|F = f).$$

Consider a security monitoring system that records whether a login originates from a previously unseen device. After investigation, each login is also classified as Normal or Attack.

The following 10 login events were observed:

Login	New Device	Classification
1	No	Normal
2	No	Normal
3	No	Normal
4	No	Normal
5	No	Attack
6	Yes	Normal
7	Yes	Attack
8	Yes	Attack
9	Yes	Attack
10	Yes	Attack

Let C represent the Classification and F represent whether the login originated from a New Device.

1. Separate the observations into the two groups $F = \text{No}$ and $F = \text{Yes}$.
2. For $F = \text{No}$, determine the probabilities of Normal and Attack and calculate

$$H(C|F = \text{No}).$$

3. For $F = \text{Yes}$, determine the probabilities of Normal and Attack and calculate

$$H(C|F = \text{Yes}).$$

4. Determine $p(F = \text{No})$ and $p(F = \text{Yes})$.
5. Calculate the overall conditional entropy

$$H(C|F) = p(F = \text{No})H(C|F = \text{No}) + p(F = \text{Yes})H(C|F = \text{Yes}).$$

6. Calculate the entropy $H(C)$ without knowing whether the login came from a new device.
7. Compare $H(C)$ and $H(C|F)$. Explain what this tells the security analyst about the usefulness of knowing whether a login originated from a new device.

1.0.7 Problem 7: Information-Theoretic Features and Anomaly Detection

In the previous problems, information-theoretic metrics were calculated individually from cyber security data. These metrics can also be combined to form a feature vector that describes the behavior of a system during a particular observation period.

For example, assume that the following three information-theoretic features have been calculated from network activity:

H_P = entropy of destination-port activity,

D_{KL} = KL divergence from the normal destination-port distribution,

and

H_A = entropy of authentication activity.

These values can be placed into a vector:

$$\mathbf{x} = [H_P, D_{KL}, H_A].$$

A feature vector provides a compact numerical description of the observed network behavior.

Cosine Similarity

One way to compare two feature vectors is cosine similarity.

For two vectors $\mathbf{x}$ and $\mathbf{y}$,

$$\cos(\theta) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

The numerator is the dot product:

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

The magnitude of a vector is

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

For vectors containing non-negative features such as those used in this problem, cosine similarity will range from 0 to 1. A value close to 1 indicates that the vectors point in similar directions, while a value closer to 0 indicates that their patterns are less similar.

Consider the simple vectors

$$\mathbf{a} = [1, 1]$$

and

$$\mathbf{b} = [2, 2].$$

Although their magnitudes are different, they point in exactly the same direction. Their cosine similarity is therefore

$$\cos(\theta) = 1.$$

This illustrates an important property of cosine similarity: it primarily compares the direction or pattern of the features rather than their absolute magnitudes.

Cyber Security Scenario

A security analyst has established the following information-theoretic feature vector from a period of normal network operation:

$$\mathbf{N} = [1.0, 0.10, 1.0].$$

The components represent

$$[H_P, D_{KL}, H_A].$$

During two later monitoring periods, the following feature vectors are obtained:

$$\mathbf{A} = [0.9, 0.20, 0.8]$$

and

$$\mathbf{B} = [0.4, 1.20, 0.3].$$

1. Calculate the dot product

$$\mathbf{N} \cdot \mathbf{A}.$$

2. Calculate the magnitudes

$$\|\mathbf{N}\|$$

and

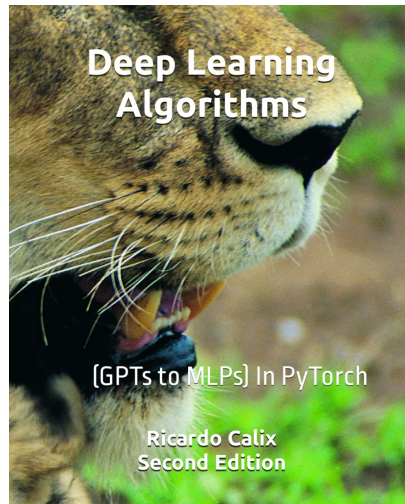
$$\|\mathbf{A}\|.$$

3. Calculate the cosine similarity between $\mathbf{N}$ and $\mathbf{A}$.
4. Repeat the calculation for $\mathbf{N}$ and $\mathbf{B}$.
5. Which monitoring period is more similar to the normal baseline?
6. Which monitoring period would be more interesting to a security analyst looking for anomalous behavior?
7. Explain why cosine similarity alone should not be considered proof that an attack has occurred.

1.1 Summary

Mastering these problems is fundamental to mastering Information Theoretic metrics for cyber security.

Other Books by Ricardo A. Calix



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